

2.1 - 2.2 Sets and Set Operations

Tuesday, October 1, 2024 10:11 AM

If $A = \{a, b, c\}$,

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

Definition: For sets A and B , the

Cartesian product of A and B is the set

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

So you make tuples like with (x, y, z) in 3-space.

Extends to products of any number of sets.

$$(A \times B) \times C \neq A \times B \times C$$

(a, b, c) vs. (a, b, c)
pt, time 3d-point

Definition: Let $P(x)$ be a propositional

function with domain D . The truth set

of $P(x)$ is the set $\{x \in D \mid P(x)\}$.

What is the truth value of

$$\exists! x \in \mathbb{Z} (x^2 - 2 = 0)$$

$$\exists! x \in \mathbb{R} (x^2 - 2 = 0)$$

$$\exists! x \in \mathbb{R}^+ (x^2 - 2 = 0)$$

What elements are in the

truth set of $\{x \in \mathbb{R} \mid x^2 - 2 = 0\}$

$$P(x): x^2 - 2 = 0 \quad D = \mathbb{R}$$

2.2 Set Operations

Def: The union of two sets A and

B is the set containing the elements

in A , or B (or both). Denoted

$$A \cup B$$

Def: The intersection of two sets A

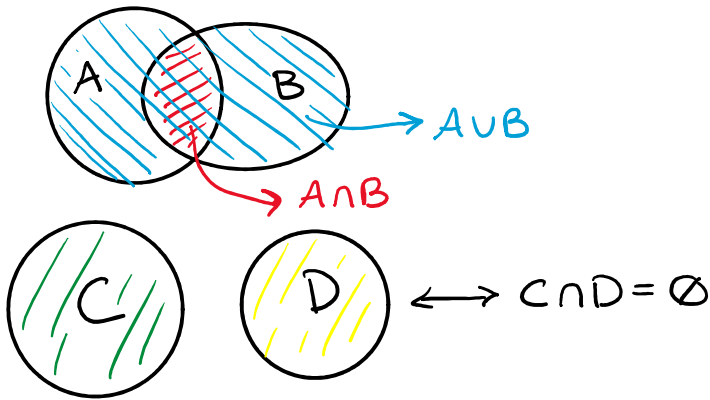
and B is the set containing the

elements that are in both A and B .

and B is the set containing the elements that are in both A and B .

Denoted $A \cap B$.

Def: Two sets are disjoint if their intersection is the empty set. ($A \cap B = \emptyset$)



Note: For finite sets A and B

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$\text{Let } A = \{\emptyset, 1, 2, a, b\}$$

$$B = \{\emptyset, 3, 4, c, d\}$$

What are $A \cup B$ and $A \cap B$?

Are A ; B disjoint?

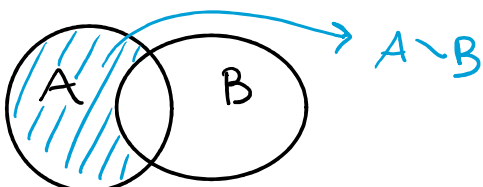
$$A \cup B = \{\emptyset, 1, 2, 3, 4, a, b, c, d\}$$

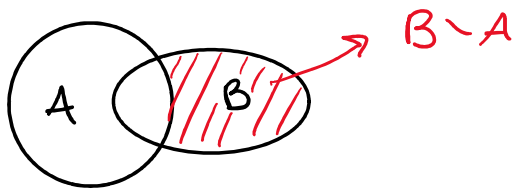
$$A \cap B = \{\emptyset\} \text{ is not empty.}$$

A ; B are not disjoint.

What about $\{1, 2, 3\} \cup \{1, 2, 3, 4\} = ?$

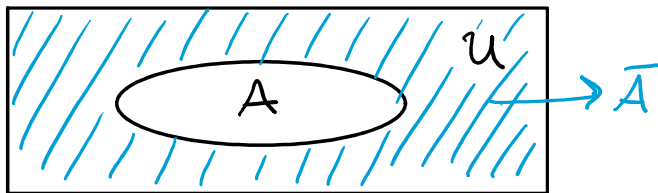
Definition: The difference of two sets A and B is denoted by $A - B$ or $A \setminus B$ and is the set containing the elements in A which are not in B . (Order matters)





Definition: A universal set U is defined to be the superset of all sets.

Definition: Let U be the universal set. The complement of A (in U) is denoted by $\bar{A}$ or A^c , and the difference of U and A ($\bar{A} = U \setminus A$).



Set-Builder Descriptions

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

$$A \setminus B = \{x \mid (x \in A) \wedge (x \notin B)\}$$

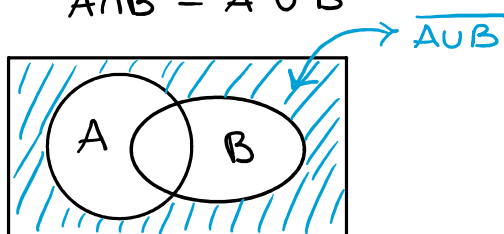
$$\bar{A} = \{x \mid (x \in U) \wedge (x \notin A)\}$$

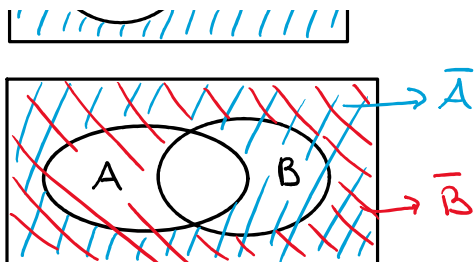
[There is a massive table of useful identities on page 136.]

De Morgan's Law of Complements

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$





Just like with conjunctions and disjunctions, there is no required order for unions and intersections.

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

Def: The union of a collection of sets $A_1, \dots, A_n$ is the set that contains elements that are in at least one of the sets $A_1, \dots, A_n$.

$$\text{Denoted } \bigcup_{i=1}^n A_i = A_1 \cup \dots \cup A_n$$

Def: The intersection of a collection of sets $A_1, \dots, A_n$ is the set that contains elements which are in every set $A_1, \dots, A_n$.

$$\text{Denoted } \bigcap_{i=1}^n A_i = A_1 \cap \dots \cap A_n$$

We even allow infinite unions and intersections.

If $A_i = \{0, 2, 4, \dots, 2i\}$ for $i = 0, 1, 2, \dots$

then find

$$\bigcup_{i=0}^{\infty} A_i = \bigcup_{i=0}^{\infty} \{0, 2, \dots, 2i\} = \{0, 2, 4, 6, \dots\}$$

and

$$\bigcap_{i=0}^{\infty} A_i = \bigcap_{i=0}^{\infty} \{0, 2, \dots, 2i\} = \{0\}$$

Show $\overline{A \cup B} = \bar{A} \cap \bar{B}$ for sets A, B

$$\overline{A \cup B} = \{x \mid (x \in U) \wedge (x \notin (A \cup B))\}$$

$$\begin{aligned}
 & \text{if } x \notin (A \cup B) \Rightarrow (x \notin A) \wedge (x \notin B) \\
 & = \{x \mid (x \in U) \wedge (x \notin A) \wedge (x \notin B)\} \\
 & = \{x \mid \underbrace{(x \in U) \wedge (x \notin A)}_{\text{same as } \overline{A}} \wedge \underbrace{(x \in U) \wedge (x \notin B)}_{\text{same as } \overline{B}}\} \\
 & = \overline{A} \cap \overline{B}
 \end{aligned}$$

Show that for sets A, B, C

$$(A \setminus C) \cap (C \setminus B) = \emptyset$$

$$A \setminus C = \{x \mid (x \in A) \wedge (x \notin C)\}$$

$$C \setminus B = \{x \mid (x \in C) \wedge (x \notin B)\}$$

$$\therefore (A \setminus C) \cap (C \setminus B)$$

$$= \{x \mid (x \in A) \wedge \underbrace{(x \notin C) \wedge (x \in C)}_{\text{contradiction}} \wedge (x \notin B)\}$$

There are no elements x that
can satisfy $x \in C$ and $x \notin C$

Therefore this set contains no elements i.e. is $\emptyset$.

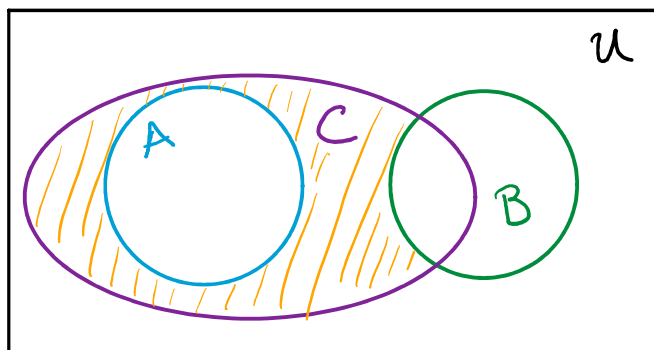
Show that if A, B, C are sets

$$\text{then } A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

$$B \cap C = \{x \mid (x \in B) \wedge (x \in C)\}$$

$$\begin{aligned}
 A \cup (B \cap C) &= \{x \mid (x \in A) \vee ((x \in B) \wedge (x \in C))\} \\
 &= \{x \mid ((x \in A) \vee (x \in B)) \wedge ((x \in A) \vee (x \in C))\} \\
 &= (A \cup B) \cap (A \cup C)
 \end{aligned}$$

Draw a Venn diagram of the sets
 U, A, B, C if U is the universal set
and $A \cap B = \emptyset$, $A \subseteq C$, $C \cap B \neq \emptyset$
 $C \setminus A \neq \emptyset$



Shade the region corresponding to
 $\bar{A} \cap (C - B)$ 